

Question-1

```
mysql> SELECT COUNT(emp_name) AS total_employees
-> FROM employee
-> ;

+-----+
| total_employees |
+-----+
|              18 |
+-----+
1 row in set (0.01 sec)

mysql> |
```

1.

2.

ERROR 1064 (42000): Unknown column 'emp_dept' in 'group statement'

```
mysql> SELECT emp_dept, MAX(income) AS max_income  
-> FROM employee  
-> GROUP BY emp_dept;
```

emp_dept	max_income
finance	280000
HR	120000
management	500000
sales	30000
accounts	800000

5 rows in set (0.01 sec)

```
mysql> SELECT MAX(income) AS max_income_overall  
-> FROM employee  
-> ;
```

max_income_overall
800000

1 row in set (0.00 sec)

```
mysql> |
```

3.

```
mysql> SELECT emp_name,income
-> FROM employee
-> WHERE income BETWEEN 100000 AND 500000
-> AND income != 120000
-> ;
```

emp_name	income
peter	100000
donalad	100000
obama	500000
mac	280000
peter	100000
donald	100000
obama	500000
mac	280000

8 rows in set (0.01 sec)

```
mysql> |
```

4.

```
mysql> SELECT COUNT(*) AS count_employees_above_1_lakh
-> FROM employee
-> WHERE income>100000
-> ;
```

count_employees_above_1_lakh
8

1 row in set (0.00 sec)

```
mysql> |
```

5.

```
mysql> SELECT emp_name, income  
-> FROM employee  
-> ORDER BY income ASC  
-> ;
```

emp_name	income
linkion	25000
linkion	25000
kane	30000
kane	30000
adam	54000
adam	54000
peter	100000
donalad	100000
peter	100000
donald	100000
mark	120000
mark	120000
mac	280000
mac	280000
obama	500000
obama	500000
manas	600000
vasin	800000

```
18 rows in set (0.01 sec)
```

```
mysql> SELECT emp_dept, COUNT(*) AS num_employees, MAX(income) AS max_income
-> FROM employee
-> GROUP BY emp_dept;
```

emp_dept	num_employees	max_income
finance	5	280000
HR	4	120000
management	4	500000
sales	2	30000
accounts	3	800000

5 rows in set (0.00 sec)

7.

```
mysql> SELECT place, COUNT(*) AS num_employees
-> FROM employee
-> GROUP BY place
-> ;
```

place	num_employees
newyork	2
california	4
arizona	1
florida	4
georgia	2
alaska	2
india	2
arizona	1

8 rows in set (0.00 sec)

8.

```
mysql> SELECT place, COUNT(*) AS num_employees
-> FROM employee
-> GROUP BY place
-> ORDER BY num_employees DESC
-> ;
```

place	num_employees
california	4
florida	4
newyork	2
georgia	2
alaska	2
india	2
arizona	1
arizona	1

8 rows in set (0.00 sec)

```
mysql> |
```

9.

```
mysql> SELECT place, COUNT(*) AS num_employees
-> FROM employee
-> GROUP BY place
-> HAVING COUNT(*)>1;
```

place	num_employees
newyork	2
california	4
florida	4
georgia	2
alaska	2
india	2

6 rows in set (0.00 sec)

```
mysql> |
```

10.

```
mysql> SELECT place, COUNT(*) AS num_employees  
-> FROM employee  
-> WHERE place != 'california'  
-> GROUP BY place  
-> HAVING COUNT(*)>=2  
-> ORDER BY num_employees DESC  
-> ;
```

place	num_employees
florida	4
newyork	2
georgia	2
alaska	2
india	2

5 rows in set (0.00 sec)

```
mysql> |
```

Question -2

1.

```
mysql> SELECT *
-> FROM BRANCH;
```

branch_name	branch_city	assets
Brighton	Brooklyn	7100000.00
Downtown	Brooklyn	9000000.00
Mianus	Horseneck	400000.00
North Town	Rye	3700000.00
Perryridge	Horseneck	1700000.00
Pownal	Bennington	300000.00
Redwood	Palo Alto	2100000.00
Round Hill	Horseneck	8000000.00

8 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM LOAN;
```

loan_number	branch_name	amount
L-11	Round Hill	900
L-14	Downtown	1500
L-15	Perryridge	1500
L-16	Perryridge	1300
L-17	Downtown	1000
L-23	Redwood	2000
L-93	Mianus	500

7 rows in set (0.00 sec)

```
mysql> SELECT *
-> FROM ACCOUNT;
```

account_number	branch_name	balance
A-101	Downtown	500.00
A-102	Perryridge	400.00
A-201	Brighton	900.00
A-215	Mianus	700.00
A-217	Brighton	750.00


```

| A-201      | Brighton | 900.00 |
| A-215      | Mianus   | 700.00 |
| A-217      | Brighton | 750.00 |
| A-222      | Redwood  | 700.00 |
| A-305      | Round Hill | 350.00 |
+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> SELECT *
-> FROM CUSTOMER;
+-----+-----+-----+
| customer_name | customer_street | customer_city |
+-----+-----+-----+
| Adams         | Spring          | Pittsfield    |
| Brooks        | Senator         | Brooklyn      |
| Curry         | North           | Rye            |
| Glenn         | Sand Hill       | Woodside      |
| Green         | Walnut          | Stamford      |
| Hayes         | Main            | Harrison      |
| Jackson       | Park            | Pittsfield    |
| Johnson       | Alma            | Palo Alto     |
| Jones         | Main            | Harrison      |
| Lindsay       | Park            | Pittsfield    |
| Smith         | North           | Rye            |
| Turner        | Putnam          | Stamford      |
| Williams      | Nassau          | Princeton     |
+-----+-----+-----+
13 rows in set (0.00 sec)

mysql> SELECT *
-> FROM DEPOSITOR;
+-----+-----+
| customer_name | account_number |
+-----+-----+
| Johnson       | A-101          |
| Hayes         | A-102          |
| Johnson       | A-201          |
| Smith         | A-215          |
| Jones         | A-217          |
| Lindsay       | A-222          |
| Turner        | A-305          |
+-----+-----+

```

2.

```

mysql> SELECT CUSTOMER.customer_name, BORROWER.loan_number, LOAN.amount
-> FROM CUSTOMER
-> JOIN BORROWER ON CUSTOMER.customer_name = BORROWER.customer_name
-> JOIN LOAN ON BORROWER.loan_number = LOAN.loan_number;
+-----+-----+-----+
| customer_name | loan_number | amount |
+-----+-----+-----+
| Smith         | L-11       | 900    |
| Jackson       | L-14       | 1500   |
| Hayes         | L-15       | 1500   |
| Adams         | L-16       | 1300   |
| Jones         | L-17       | 1000   |
| Williams      | L-17       | 1000   |
| Smith         | L-23       | 2000   |
| Curry         | L-93       | 500    |
+-----+-----+-----+
8 rows in set (0.00 sec)

```

3.

```
mysql> SELECT CUSTOMER.customer_name, BORROWER.loan_number, LOAN.amount
-> FROM CUSTOMER
-> JOIN BORROWER ON CUSTOMER.customer_name = BORROWER.customer_name
-> JOIN LOAN ON BORROWER.loan_number = LOAN.loan_number;
```

customer_name	loan_number	amount
Smith	L-11	900
Jackson	L-14	1500
Hayes	L-15	1500
Adams	L-16	1300
Jones	L-17	1000
Williams	L-17	1000
Smith	L-23	2000
Curry	L-93	500

8 rows in set (0.00 sec)

4 and 5

```
mysql> SELECT DISTINCT b1.branch_name
-> FROM BRANCH b1
-> WHERE b1.assets > (
->     SELECT MIN(b2.assets)
->     FROM BRANCH b2
->     WHERE b2.branch_city = 'Brooklyn'
-> );
```

branch_name
Downtown
Round Hill

2 rows in set (0.00 sec)

```
mysql> SELECT c.customer_name
-> FROM CUSTOMER c
-> JOIN BORROWER b ON c.customer_name = b.customer_name
-> JOIN LOAN l ON b.loan_number = l.loan_number
-> WHERE l.branch_name = 'Perryridge'
-> ORDER BY c.customer_name ASC;
```

customer_name
Adams
Hayes

2 rows in set (0.00 sec)

6,7,8

```
mysql> SELECT *  
-> FROM LOAN  
-> ORDER BY amount DESC, loan_number ASC;
```

loan_number	branch_name	amount
L-23	Redwood	2000
L-14	Downtown	1500
L-15	Perryridge	1500
L-16	Perryridge	1300
L-17	Downtown	1000
L-11	Round Hill	900
L-93	Mianus	500

7 rows in set (0.00 sec)

```
mysql> SELECT AVG(balance) AS avg_balance  
-> FROM ACCOUNT;
```

avg_balance
614.285714

1 row in set (0.00 sec)

```
mysql> SELECT COUNT(*) AS num_tuples  
-> FROM CUSTOMER;
```

num_tuples
13

1 row in set (0.00 sec)

9,10,11,12

```
mysql> SELECT SUM(amount) AS total_loan_amount
-> FROM LOAN;
+-----+
| total_loan_amount |
+-----+
|          8700 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT AVG(balance) AS avg_balance
-> FROM ACCOUNT
-> WHERE branch_name = 'Perryridge';
+-----+
| avg_balance |
+-----+
| 400.000000 |
+-----+
1 row in set (0.00 sec)

mysql> SELECT branch_name, AVG(balance) AS avg_balance
-> FROM ACCOUNT
-> GROUP BY branch_name;
+-----+-----+
| branch_name | avg_balance |
+-----+-----+
| Brighton   | 825.000000 |
| Downtown   | 500.000000 |
| Mianus      | 700.000000 |
| Perryridge  | 400.000000 |
| Redwood    | 700.000000 |
| Round Hill  | 350.000000 |
+-----+-----+
6 rows in set (0.00 sec)

mysql> SELECT branch_name, AVG(balance) AS avg_balance
-> FROM ACCOUNT
-> WHERE balance > 1200
-> GROUP BY branch_name;
Empty set (0.00 sec)
```

```
mysql>
mysql> SELECT a.branch_name, COUNT(DISTINCT d.customer_name) AS num_depositors
-> FROM DEPOSITOR d
-> JOIN ACCOUNT a ON d.account_number = a.account_number
-> GROUP BY a.branch_name;
```

branch_name	num_depositors
Brighton	2
Downtown	1
Mianus	1
Perryridge	1
Redwood	1
Round Hill	1

6 rows in set (0.00 sec)

```
mysql> SELECT d.customer_name, AVG(a.balance) AS avg_balance
-> FROM DEPOSITOR d
-> JOIN ACCOUNT a ON d.account_number = a.account_number
-> JOIN CUSTOMER c ON d.customer_name = c.customer_name
-> WHERE c.customer_city = 'Harrison'
-> GROUP BY d.customer_name
-> HAVING COUNT(d.account_number) >= 3;
```

Empty set (0.00 sec)